Cutting Telecom Churn with “Predict → Time → Offer”

A data-driven retention proposal built on 100,000 customers

GCI World 2026 September – Final Assignment

Executive summary

49.6%

of the 100,000 customers churn – retention is the core business problem

56% vs 39%

churn when handset is older vs younger than ~300 days

AUC 0.69

gradient-boosting model; top risk decile churns 78% vs 20% in the lowest

- Customers leave when the handset ages past ~300 days and around months 11–13 of tenure.
- Proposal: score customers monthly, act before the cliff, and send a timed handset-upgrade / loyalty offer.
- Targeting the top 70% by risk earns ≈ $672K per cycle (+$124K vs contacting everyone) under stated assumptions.
- Next step: an A/B pilot to measure the real take-up rate.

Client & market context

- Client: a mobile operator – monthly revenue per customer ≈ $59, 100,000 customers in the data.
- Mobile markets are saturated: growth depends on keeping subscribers rather than finding new ones.
- Harvard Business Review [1]: acquiring a customer can cost 5–25x more than retaining one (varies by study and industry), so early, targeted retention is a standard focus for operators.
- Common practice: mass discounts or last-minute retention calls when the customer already wants to cancel – costly and late.
- Question for this project: who is about to leave, when, and what should we offer?

Data & approach

100,000

customers; Client.csv + Record.csv joined on Customer_ID (100 columns)

churn

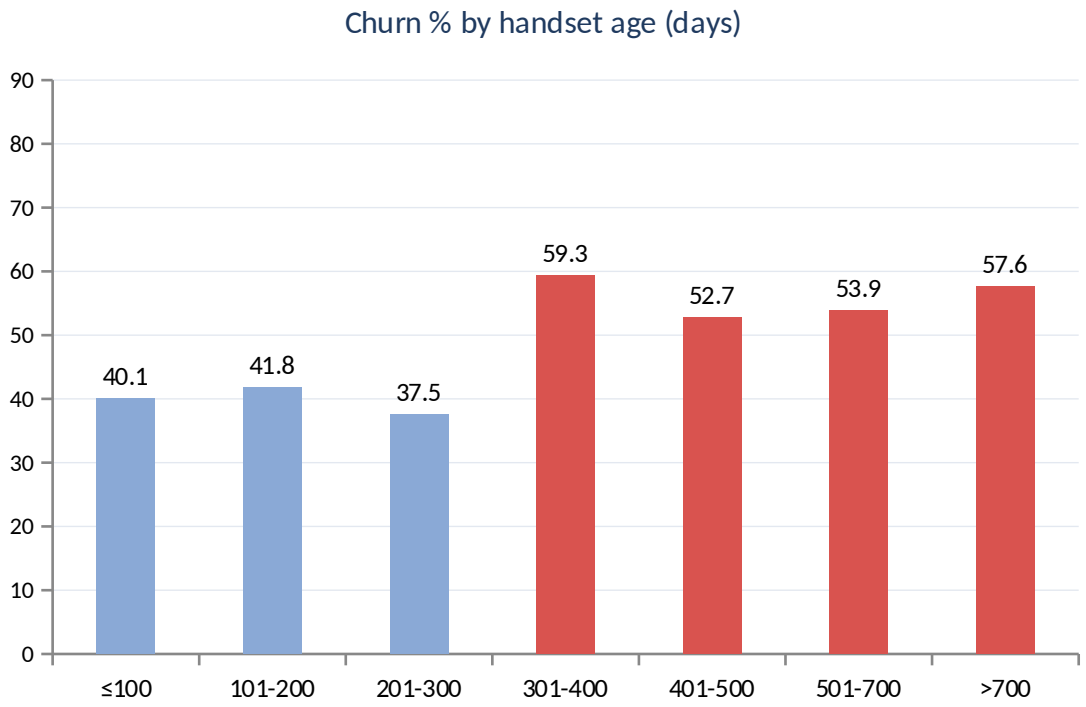
target: 1 = customer left (≈ 50% of rows, so data is likely re-sampled)

6 new

features: usage trend, drop/block rates, revenue per minute, handset age >300d …

- Workflow: understand data → EDA → hypothesis → model → proposal → verification.
- Missing values (mostly demographics) kept and handled in the pipeline instead of dropping customers.
- 20% hold-out test set, stratified; all numbers in these slides come from the test set or the full data.

EDA 1 – Churn jumps once the handset is older than ~300 days



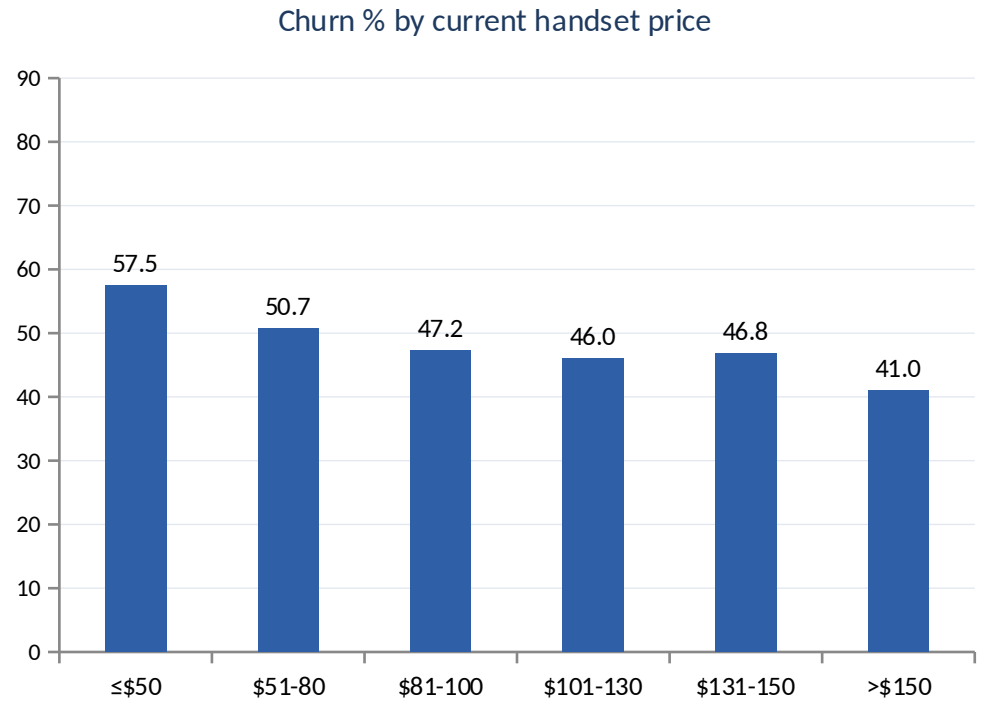
56.3%

churn if handset >300 days (60% of customers)

39.5%

churn if handset ≤300 days

EDA 2 – Tenure cliff at months 11–13 and low-value handsets



61.4%

churn at months 11–13 (19,713 customers) – likely end of first commitment

33.8%

churn at months 6–10 – the safest group

Frequent customer-care callers churn less (42.9% vs 51.8% for none); falling usage is an early warning.

Hypothesis & bottleneck

Hypothesis

Churn is triggered when the handset ages past ~300 days / the first ~12-month commitment ends – especially with low-value handsets. If the operator identifies these customers in advance and offers a timed upgrade, churn in the targeted group can be reduced profitably.

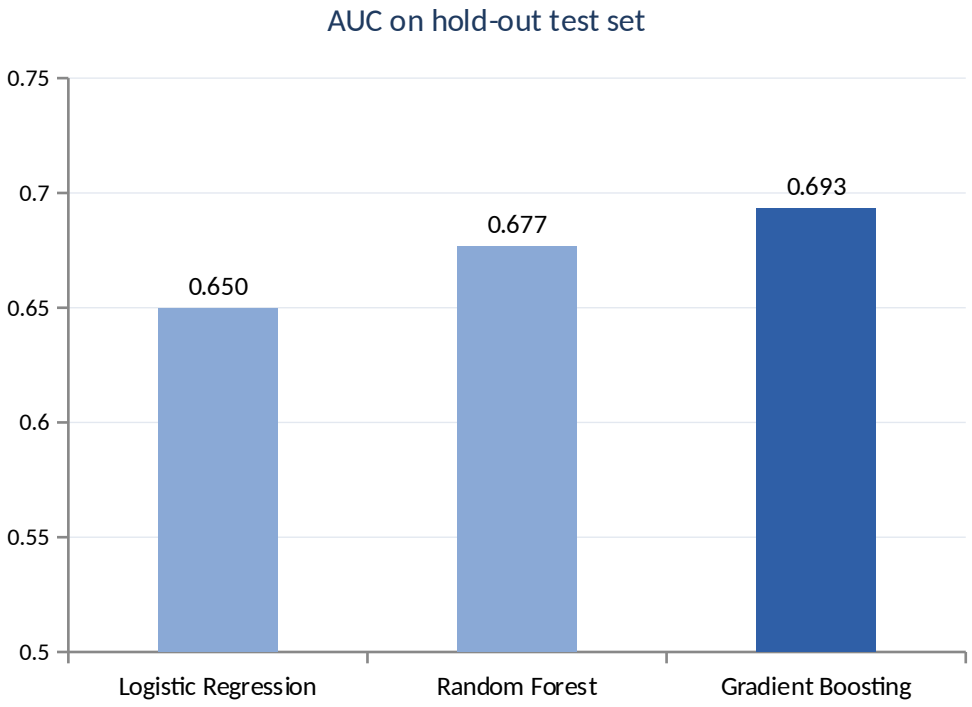
Bottleneck

Retention teams react too late and without a risk ranking, so offers are spent on customers who would have stayed.

Stakeholders

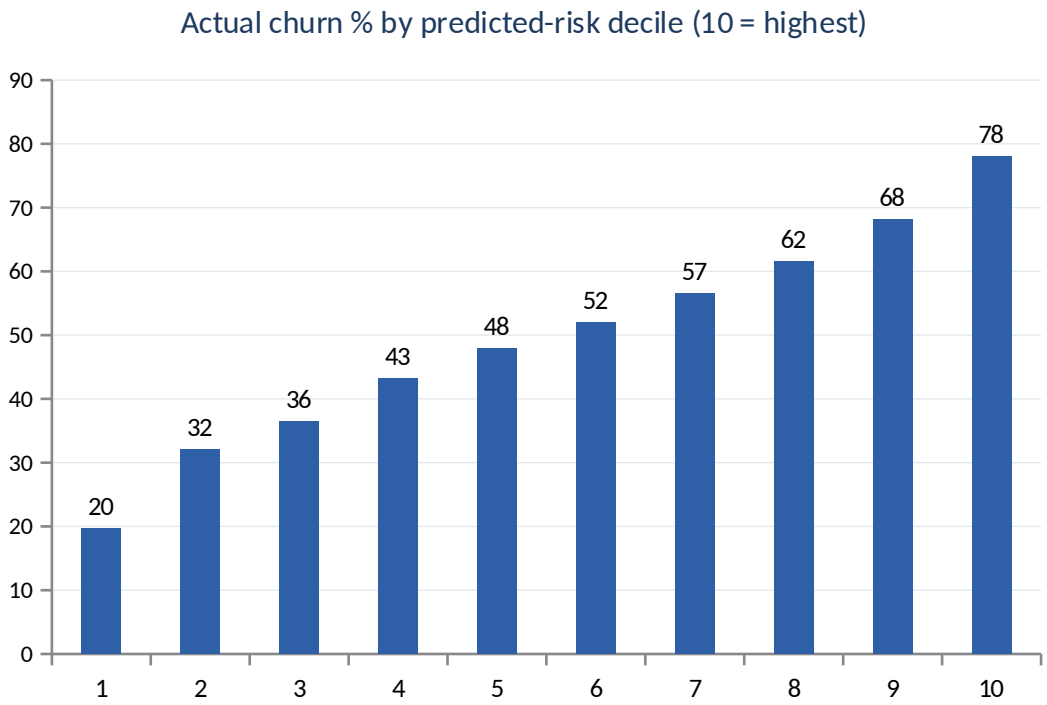
CRM / retention team, sales & device procurement, finance, customers.

Models – gradient boosting ranks churn risk best



- Logistic regression, random forest and gradient boosting compared on the same split.
- Gradient boosting: AUC 0.693, accuracy 63.8%, F1 0.641.
- AUC ≈ 0.69 is realistic for behavioural churn data; the value lies in ranking customers, shown next.

The model separates safe from at-risk customers



Top drivers

- Tenure (months)
- Handset age (eqpdays)
- Change in minutes of use
- Monthly recurring charge
- Handset price

Proposal – Predict → Time → Offer

1 Predict

Score every active customer monthly with the gradient-boosting model.

2 Time

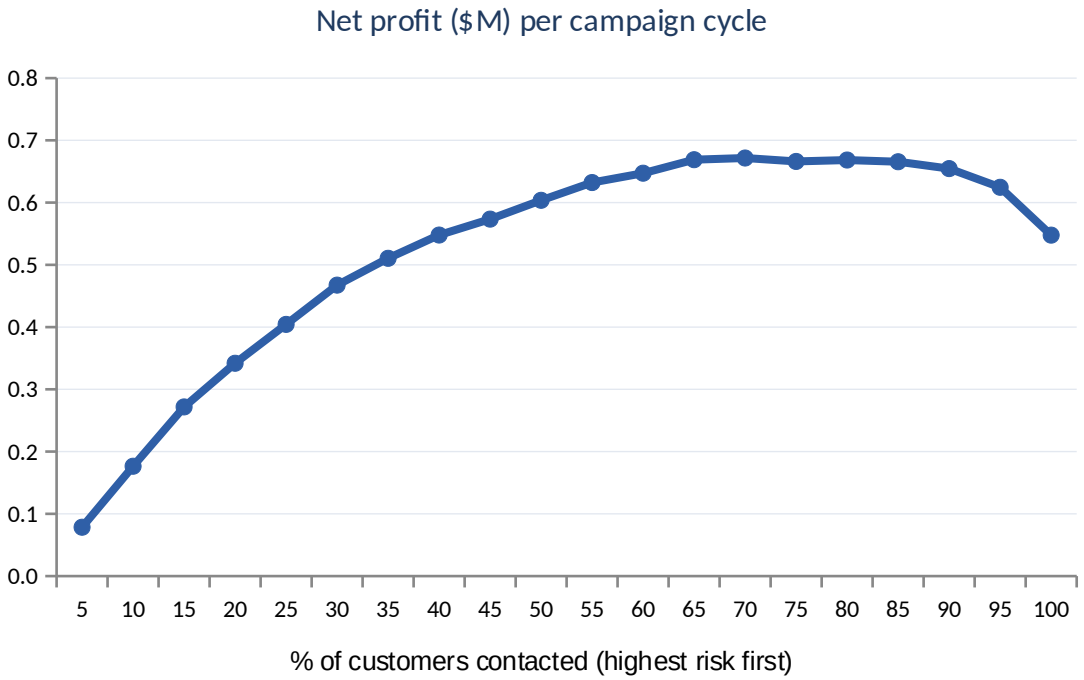
Trigger the campaign for high-risk customers approaching handset age ~250–300 days or tenure ~10–11 months – before the cliff.

3 Offer

Handset-upgrade or loyalty offer, prioritised by risk and customer value, within a fixed budget.

Success metric: net campaign profit = saved-customer value − contact and incentive costs.

Verification – profit by share of customers contacted



$672K

best case: contact top 70% by risk

$548K

contact everyone (no model)

Assumptions (not from the data): contact $2, incentive $100 paid on acceptance, 20% take-up, 40% margin, 12 months kept. Simulated on actual test outcomes, scaled to 100k customers.

Robustness, risks and next steps

Max profit ($M)	take-up 10%	take-up 20%	take-up 30%
incentive $60	0.60	1.38	2.17
incentive $100	0.27	0.67	1.08
incentive $140	0.08	0.23	0.38

All 9 tested scenarios stay positive, but range from ≈ $0.08M to $2.2M – the real take-up rate decides the outcome.

Risks & next steps

- Pilot: A/B test targeted vs control group to measure real take-up.
- Dataset churn ≈ 50% looks re-sampled; recalibrate on the live base.
- No contract-type or competitor data; costs and margin are assumptions.
- Retrain monthly and monitor drift.

References

- Dataset: Client.csv and Record.csv, GCI World course (Matsuo-Iwasawa Lab, UTokyo).
- scikit-learn documentation (HistGradientBoostingClassifier, permutation_importance): https://scikit-learn.org/stable/
- [1] The Value of Keeping the Right Customers, Harvard Business Review, 29 Oct 2014. https://hbr.org/2014/10/the-value-of-keeping-the-right-customers
- Generative AI (Claude) used for analysis and code support – chat-history URL is listed in the submission form.